12 **(a)** Show that for the point on the curve where $x = 0$, $\frac{3a-x}{a+x} \geqslant 0$ as given in line 10. **[1]**

(b) Use the equation of the curve given in line 8 to show that for all points on the curve where $x \neq 0$, $\frac{3a-x}{a+x} \geqslant 0$ as given in line 10. **[1]**

(c) Show that it follows that $3a-x \geqslant 0$ and $a+x > 0$, as given in lines 10 and 11, for values of $a > 0$. **[3]**

(d) Hence find, in terms of a where $a > 0$, the range of values of x for points on the curve $y^2 = \frac{x^2(3a-x)}{a+x}$. **[1]**

(e) Find the equation of the asymptote to the curve. **[1]**